Report On Amazon Retail Analytics

Business Problem

As a data analyst, I conducted an end-to-end analytics project using a publicly available delivery operations dataset obtained from Kaggle. The dataset simulates real-world last-mile logistics operations of Amazon or other e-commerce delivery platforms.

In last-mile delivery services, one of the most critical operational challenges is ensuring that orders reach customers within the promised time. Late deliveries lead to customer dissatisfaction, negative ratings, refunds, and increased operational costs. Despite having large volumes of operational data, organizations often struggle to identify the exact causes of delivery delays because multiple factors may simultaneously affect delivery performance.

The business lacked clear visibility into:

- Which operational factors are responsible for delivery delays
- Whether traffic, weather, or delivery distance significantly impacts delivery time
- If workload distribution among delivery partners affects efficiency
- Whether demographic characteristics of delivery partners influence performance

The goal of this project was to analyze delivery operations data and transform raw operational records into meaningful business insights. By identifying the main drivers of delivery delays, the analysis aims to help logistics managers make data-driven decisions such as optimizing delivery partner allocation, improving route planning, and setting realistic delivery time expectations for customers.

Tools Used and Analytical Workflow

This project was completed using a structured data analytics workflow. Each tool was selected for a specific purpose in the data analysis pipeline.

Python (Pandas, NumPy, Matplotlib)

Purpose: Data Cleaning and Preparation

Python was used as the first step after collecting the dataset from Kaggle. The raw dataset contained missing values, inconsistent records, and outliers that could affect analysis.

Python was used to:

- Inspect and understand the dataset structure
- Handle missing values
- Remove incorrect and duplicate records
- Calculate delivery distance using latitude and longitude
- Create new features such as rating categories and performance indicators
- Detect and remove unrealistic distance outlier

SQL Server

Purpose: Data Querying and Business Analysis

After cleaning the data, the processed dataset was imported into SQL Server. SQL was used to simulate a real company database environment and to perform structured data analysis.

SQL was used to:

- Count total orders
- Calculate average delivery time
- Analyze delivery time by traffic, weather, and city
- Evaluate the impact of delivery partner ratings
- Study workload (multiple deliveries) impact on delays

Power BI

Purpose: Data Visualization and Dashboard Development

After analysis, Power BI was used to convert analytical results into an interactive business dashboard. The dashboard presents insights in a visual format understandable by business managers.

Power BI was used to:

- Create KPI cards (orders, delivery time, distance, ratings)
- Visualize traffic and weather impact on delivery time
- Compare partner performance based on ratings
- Build a scatter plot to evaluate age vs performance
- Add interactive filters (city slicer) and navigation buttons

Excel

Purpose: Initial Data Inspection

Excel was used only for a quick initial inspection of the dataset to view columns and understand basic data structure before importing into Python.

Methodology

Python

- Loaded the Kaggle dataset into Pandas DataFrame
- Checked missing values and column types
- Removed unnecessary columns and incorrect records
- Converted columns into proper numeric formats
- Created new useful variables

Example tasks performed:

- Filled missing delivery partner ratings with average value
- Removed invalid records where delivery distance was unrealistic
- Calculated delivery distance using restaurant and customer latitude/longitude
- Created delivery speed categories (Fast, Normal, Slow)

Raw data cannot be analyzed directly. Cleaning and preparing the data ensured accurate results in later analysis.

SQL Server

- Created a database and imported the processed dataset
- Wrote SQL queries to answer business questions

Example queries performed:

- **Counted total number of orders**

	Total_Orders
1	2422

- **Calculated average delivery time**

	Avg_Delivery_Time
1	26.5012386457473

- **Compared delivery time across traffic levels**

	Road_traffic_density	Avg_Time
1	1	31.078104993598
2	0	28.0524193548387
3	3	26.4454225352113
4	5	24.64
5	2	21.65

- **Checked performance based on delivery partner ratings**

	Rating_Category	Avg_Delivery_Time
1	Excellent	24.3352012226184
2	Good	35.539641943734
3	Average	37.0307692307692
4	Poor	37.6666666666667

- **Analyzed impact of multiple deliveries on delays**

	multiple_deliveries	Avg_Time
1	3	48.5
2	2	40.6037735849057
3	1	26.8235294117647
4	0	23.1367521367521

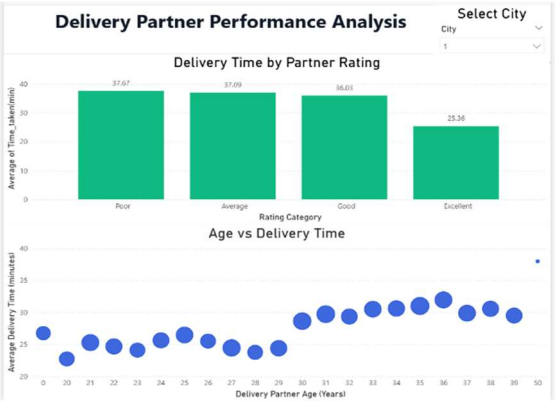
Comment:

The SQL analysis revealed that delivery times increased significantly during heavy traffic conditions. In real-world companies, analysts regularly work with databases, and SQL helps them efficiently organize, analyze, and summarize large volumes of data to generate useful insights quickly.

Power BI – Visualization & Dashboard

What I built:

- KPI cards showing total orders, average delivery time, average distance, and ratings
- Traffic vs delivery time chart
- Weather impact chart
- Delivery partner rating performance chart
- Scatter plot for age vs delivery time
- Interactive city filter (slicer)
- Page navigation buttons



Comment:
Managers can quickly understand that delivery times tend to rise when traffic is heavy or when orders are placed over longer distances. However, SQL tables are not always easy for managers to interpret. A dashboard helps by transforming the analysis into simple visuals, making it easier for them to understand the data and make clear decisions.

Excel – Initial Data Understanding

- Opened dataset to view column names
- Checked data format and structure
- Verified files before importing into Pythons

Analytical Results and Business Findings

The objective of this project was to identify the operational factors responsible for delivery delays and evaluate whether delivery performance depends on environmental conditions, workload, or delivery partner characteristics.

After performing Python preprocessing, SQL analysis, and Power BI visualization, several clear patterns were identified

Overall Delivery Performance

- Average delivery time ≈ 26 minutes
- Delivery time varied significantly across orders
- Some deliveries were completed quickly, while others experienced delays

Interpretation:
The company does not have a consistent delivery performance. There are operational inefficiencies affecting completion tim

Impact of Traffic :

- Delivery time increased noticeably under higher traffic density.
- Traffic congestion is one of the primary causes of delivery delays.
- Late deliveries are not random they are strongly linked to road conditions.

Impact of Weather :

- Adverse weather conditions increased average delivery time.
- Environmental factors directly influence delivery efficiency.
- Customers are promised a fixed delivery time even when conditions are unfavorable, which creates dissatisfaction.

Impact of Delivery :

- Longer delivery distance led to longer delivery time
- Delivery time depends heavily on route distance.
- Improper partner assignment (far partner assigned to near order) is likely increasing delays.

Delivery Partner Rating :

- Higher-rated delivery partners completed deliveries faster.
- Performance varies by skill and efficiency level of delivery partners.
- Training and monitoring delivery partners can improve overall performance.

Multiple Deliveries :

- When delivery partners handled multiple orders simultaneously, delivery time increased.
- Overloading delivery partners causes delays.
- Poor workload distribution is an operational problem.

Delivery Partner Age :

- No significant relationship was found between age and delivery time.
- Demographic characteristics do not influence performance.
- Operational planning is more important than employee demographics.

Problems that Cause Delay in Delivery :

The business problem was to identify the cause of delivery delays.
The analysis clearly showed that delivery delays are caused mainly by operational factors, not random events and not employee demographics.

The main causes of delays were:

- Traffic congestion
- Long delivery distance
- Bad weather
- Multiple order workload

Therefore, the company can now take targeted actions instead of guessing the reasons for late deliveries.

Recommended Actions

Operational Improvements

- Assign delivery partners based on nearest location to the customer
- Limit the number of simultaneous deliveries per partner
- Increase number of partners during peak traffic hours
- Optimize route planning using navigation data

Policy Improvements

- Adjust estimated delivery time during heavy traffic or bad weather

- Introduce dynamic delivery time prediction instead of fixed timing
- Prioritize short-distance orders for faster service

Workforce Improvements

- Provide training for low-rated delivery partners
- Monitor partner performance using rating-based evaluation
- Reward high-performing delivery partners

Customer Experience Improvements

- Inform customers about possible delays in advance
- Show real-time tracking and updated delivery estimates
- Reduce refunds caused by unrealistic delivery promises

Conclusion Insights :

- This project analyzes delivery operations data to identify the main factors causing delivery delays in last-mile logistics services of Amazon.
- Using a structured analytics workflow, the dataset was cleaned and prepared in Python, analyzed using SQL Server, and visualized through an interactive Power BI dashboard.
- The analysis revealed that traffic congestion, delivery distance, bad weather, and multiple deliveries handled by a single partner are the primary drivers of delivery delays.
- In contrast, demographic factors such as delivery partner age showed no significant impact on delivery performance.
- The findings highlight that improving operational planning, route optimization, and workload distribution can significantly enhance delivery efficiency.
- By applying these data-driven insights, logistics companies can reduce delays, improve customer satisfaction, and optimize overall service quality.

Future Scope :

- In the future, this analysis can be improved by using larger and real-time datasets from logistics companies, which would help generate more accurate and practical insights.
- Machine learning techniques could also be applied to predict delivery times by considering factors such as traffic conditions, weather, delivery distance, and order workload.
- Companies can further improve efficiency by using advanced route optimization and real-time traffic or weather data, allowing them to assign the most efficient delivery routes and provide better delivery time estimates.
- Additionally, real-time dashboards and predictive analytics can help managers monitor delivery performance instantly, reduce delays, and improve the overall customer experience.